

Canonical source: 00-CHARTER/GUIDES/ATOMSMASHER_PRODUCTION.md

Æ Orange AI Computer AtomSmasher Production

AtomSmasher turns large project history into a small, source-backed operational workbench. It reduces repeated reading and model context without reducing the authority of cold source or receipt truth.

Production Boundary

AtomSmasher has three distinct contracts:

Contract	Requirement
Cold ledger	byte-exact, hash-verifiable, replayable truth
Operational view	compact commitments, structures, and source pointers
Model boundary	bounded task context with an explicit expansion path

The canonical implementation is `12-ATOMSMASHER/full-scope`. Older module directories remain useful primitives and cross-checks; they are not competing complete engines.

Work Pipeline

```
source, upload, code, data, or tool result
-> ingest and coverage receipt
-> order and commitment extraction
-> AIR structural frames, equations, and residuals
-> exact, semantic, and runtime cache lookup
-> sparse task workset
-> least-action route
-> expansion warrant when evidence is insufficient
-> answer, build, or action
-> saved-work certificate
-> compression-debt and learning receipts
```

Commitment atoms preserve decisions with continuing force. AIR frames preserve structure. Equations and residuals represent regular numeric material without hiding exceptions. Cartridges preserve reusable domain state. Sparse worksets hold the current task neighborhood. Expansion warrants authorize more context when the compact workbench cannot support a safe answer.

Truth And Compression

Operational compression is not deletion. Exact source remains addressable, and consequential claims must retain source pointers. When compact state causes a miss, wrong recall, repeated hydration, or downstream rework, record compression debt and add the case to evaluation.

Never quote one ratio without its contract. Byte compression of a receipt-shaped corpus, operational workbench reduction, and live model-context reduction use different denominators and answer different questions.

Current Bounded Evidence

The accepted Context Crystal held-out suite used 794 sources totaling 7,056,795 bytes across five cases. All 5/5 cases preserved required answer material and source-pointer checks. The measured operational context ratio ranged from `1,422.901x` to `1,445.487x`.

That result is workload-specific. It is not a universal codec ratio, an arbitrary prompt guarantee, or permission to discard the source corpus.

Production Gates

1. Ingest coverage identifies every expected source.
2. Cold truth remains immutable and hash-checkable.
3. Workbench outputs retain exact source pointers.
4. Held-out answer parity and pointer verification pass together.
5. Ratios name their workload and denominator.
6. Concurrency and restart behavior preserve writes and replay.
7. Compression debt remains visible.
8. Rollback can restore the prior incumbent and workbench behavior.

Operate And Verify

```
cd C:\AtomEons\Orange5
bun run test:atomsmasher
bun run bench:context-crystal-quality
bun run atomsmasher -- --help
```

Use a disposable database for exploratory CLI work. Review emitted case records, source hashes, reconstruction checks, and receipt hashes before promoting a result.

Failure Rules

- Missing source pointer: hydrate exact source and keep the workbench unproven.
- Ratio regression: inspect denominator, source coverage, and task parity before tuning compression.
- Wrong commitment: preserve the contradiction, correct from higher evidence, and add a held-out case.
- Concurrent write loss: stop promotion; a fast partial ledger is not production.
- Stale cache hit: invalidate by source identity and record the mismatch.

Related Guides

- [Memory and Learning](#)
- [Receipts and Audit](#)
- [Technical Architecture](#)
- [Proof and Benchmarks](#)